Ruijia Yang

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EDUCATION

Hong Kong University of Science and Technology (Guangzhou) | 2024-09 - Present

PhD in Data Science and Analytics. Advisor: Prof. Zeyi Wen

Sun Yat-sen University | 2020 - 2024

Bachelor of Engineering in Computer Science and Technology, Computer Architecture

RESEARCH INTERESTS

Machine learning systems; memory-efficient LLM training; heterogeneous and distributed computing.

PUBLICATIONS

EHR-Bench: A Chinese Multi-Task Benchmark for Recruitment AI

Ruijia Yang, Bing Xu, Zulong Chen, Zhipeng Xu, Jia Xu, Yipeng Yu, Ying Zhou, Zeyi Wen.

arXiv preprint, 2026 | PDF · Code · arXiv

SlideDP: Scaling Host-Resident LLM Fine-Tuning Across Multiple GPUs

Ruijia Yang, Shiyuan Lin, Yulong Ao, Zhiyu Li, Yingli Zhao, Xianduo Li, Yonghua Lin, Zeyi Wen.

arXiv preprint, 2026 | PDF · Code · arXiv

An Efficient Heterogeneous Co-Design for Fine-Tuning on a Single GPU

Ruijia Yang, Zeyi Wen.

DAC, 2026 | PDF · Publisher · Code · arXiv · DOI

Efficient Mask Learning for Language Model Fine-Tuning

Minping Chen, **Ruijia Yang**, Zeyi Wen.

CIKM, 2025 | PDF · DOI

RESEARCH EXPERIENCE

Doctoral Research. HKUST(GZ), 2024-Present. Developed SlideFormer for single-GPU full-parameter fine-tuning and SlideDP for multi-GPU scaling through heterogeneous execution and model-state offloading.

Undergraduate Research. National Supercomputing Center in Guangzhou, 2021-2023. Supervisor: Zhiguang Chen. Studied key-value storage and distributed databases, focusing on workload-specific optimization of LevelDB.

HONORS AND AWARDS

ASC 2026 Student Supercomputer Challenge (2026)
First Prize. Advisor to the HKUST(GZ) team.

ISC 2023 Student Cluster Competition (2023)
4th Place. Team Leader.

10th ASC Student Supercomputer Challenge (22-23) (2023)
First Prize and Application Innovation Award.

Tencent Scholarship (2022-2023)
First Prize. Sun Yat-sen University.

SYSU Outstanding Student Scholarship (2020-2023)
Second Prizes.

TEACHING

DSAA 2042: Computer Architecture and Systems. Graduate Teaching Assistant, HKUST(GZ), Summer 2026.

DSAA 5003: Automatic Machine Learning. Graduate Teaching Assistant, HKUST(GZ), Fall 2025.